Writing for a human reader: what five projects taught us, what exists, and a plan for humanvoice

Prepared for the humanvoice project

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Abstract

Between May and August 2026, agents in this workspace wrote five large bodies of human-facing text: funding proposals and a 526-page thesis around the BGS model, the SMEwallet north star, the BayesFilter monograph and its companion notes, the MacroFinance CIP monograph, and the ZLB discontinuous-HMC survey. Every one of them was mathematically serious, and every one of them was initially rejected by its human reader. This document collects that failure record, inventories the tools we built in response, surveys the research literature on why language models write this way, evaluates fourteen open-source projects against our needs, and proposes a concrete design for humanvoice: a consolidated home for the writing doctrine, a measurement layer that turns prose complaints into numbers, an evaluation corpus built from our own repair history, and a deliberately thin process. The single most important empirical finding from our own record is that the failures arrive in layers, each masking the next: internal register hides mechanical style tells, which hide defensive prose, which hides missing substance. The single most important negative finding is that three months of governance machinery around the BGS thesis never produced a document a human accepted. The proposal therefore spends its budget on diagnostics and reader tests, not on process.

1 Introduction

A real reader, shown a proposal that had already passed several rounds of agent review, responded: "I don't understand the purpose. I am not sure what all those filenames are for. I am not sure what we actually try to achieve. It feels very handwavy, without much substance. It feels like a governance thing without any real product substance."[1] That verdict, from June 2026, is the problem humanvoice exists to solve. Our agents produce documents whose mathematics survives audit and whose prose loses the reader before the mathematics is ever reached.

Several partial answers already exist in the workspace: a written prose policy with a precedence order, two writing skills, an installed diagnostic pattern list, source-grounding and math-audit tools, and one fully documented rescue of a real document (the ZLB survey). It also contains an unusually candid record of what did not work, including the projects' own root-cause audits of their writing failures. humanvoice has two jobs: consolidate the pieces that earned their place, and build the pieces the record shows are missing.

[1] `DynareMCP/docs/AIpostdoc/reviews/proposal_reader_rejection_failure_2026_06_14/actual_reader_failure_record.md`.

This document reports the groundwork. Section 2 reads the failure record across the five projects. Section 3 inventories what we already have. Section 4 summarizes the research literature, and Section 5 the open-source field. Section 6 gives the proposal. Internal artifacts are cited by repository path in footnotes, since for this audience the paths are the evidence; the published literature is cited conventionally.

2 What five projects taught us

Read the five efforts in sequence, because each one's cure became the next one's disease. The credit-card work taught the agents to harden their claims. The hardening culture, applied to a thesis, produced governance prose. The reaction to governance prose produced reader gates and teaching narratives. And the last project in the sequence, the ZLB survey, had to strip the accumulated habits away layer by layer, leaving the cleanest map we have of what actually stands between an agent draft and a readable document.

2.1 cardnpv: claims without discipline

What the task calls "cardnpv" is the credit-card NPV program in MathDevMCP; no directory of that name exists. It produced a component proposal in eight versions plus a final submission, a customer-NPV survey, and about fifteen plan and audit files.[2] Its failures ran in the opposite direction from everything that followed: reviewers found the drafts too loose, not too guarded. The experimental-design audit called the section "still too handwavy for a technical panel" and demanded estimands, identification assumptions, and power calculations; the survey audit found "notation collisions and a few over-strong claims" and prose that "sounds stronger than that evidence posture warrants." The first self-review of the proposal ended in "REWRITE REQUIRED" with seven material issues.

Two things date from this period. The v1–v8 rewrite trail is the workspace's first record of a document converging under repeated review. And the plain-language non-evasion policy was written the same week as these audits:[3] the demand that agents say "wrong relative to the stated target" rather than "not necessarily wrong." The hardening was justified. What nobody predicted was how the hardening culture would sound once it reached book length.

2.2 The BGS thesis: governance in place of writing

The AIpostdoc experiment in DynareMCP is the largest and most instructive failure. For three months (14 May to 12 August 2026), agents acting as an "AI postdoc" were to turn the BGS macro-finance paper into a readable thesis, survey, and funding proposals, with the mission statement naming prose as the product: the deliverable "should be easy to understand and a pleasure to read."[4] The effort generated 18,231 files, among them 5,890 markdown documents, 103 review directories, 78 reset memos, 89 templates, and 22 thesis snapshots, and ended with a 526-page volume whose release record reads `independent_reader_pending`.[5] No human acceptance of any major artifact was ever recorded.

[2] `MathDevMCP/docs/credit-card-npv-component-proposal/`, `MathDevMCP/docs/credit-card-npv-survey/`, and `MathDevMCP/docs/plans/credit-card-npv-*`.

[3] `claudecodex/docs/plans/claudecodex-plain-scientific-language-non-evasion-policy-close-2026-07-01.md`.

[4] `DynareMCP/docs/AIpostdoc/mission_control.md`, line 39.

[5] `DynareMCP/docs/AIpostdoc/finalBGS/bgs_final_release_record_2026_08_12.md`.

Nobody diagnosed the register failure more precisely than the project itself. Its writing audit describes the mechanism: "Once internal labels, row IDs, phase names, and nonclaim formulas become the working memory of the author, they leak into the prose. The reader then sees the control plane rather than the research object."[6] The quantitative capstone is the whole-document prose audit of August 2026: all 2,775 paragraphs of the 515-page volume were classified, 815 needed revision, and the issue counts form an empirical taxonomy of agent register defects: unnatural cadence 325, administrative register 276, self-reference 237, epistemic blur 144, abstract actors 108, overcompression 65, mixed metaphor 62, vague referents 61, defensive qualification 30, buried findings 18.[7] Those pairs are, as far as we know, the best evaluation corpus for this problem anywhere, and Section 6 builds on them.

Defensiveness deserves its own paragraph, because here the record shows causation, not just correlation. The project's anti-overclaim governance mandated non-claims blocks and claim-status labels; readers then rejected exactly those blocks. The persona review that blocked the proposal used the labels "dry safety substituted for honest selling" and "compliance theater substituted for proposal pull," noting that "opening/ask non-claim blocks are repeatedly named as compliance theater that drains momentum."[8] The audit later named the underlying confusion: "honest nonclaims" were being mistaken "for completed intellectual work . . . 'not established' is not an explanation." Integrity apparatus, in other words, is not neutral in the prose. Left on the page, it reads as evasion.

Hardest-won of all is the process verdict. The project's own proposal-to-behavior audit concludes: "The repository built an increasingly capable referee and traffic cop around an author whose method of inquiry and composition was largely unchanged . . . A system can block premature release and still never produce something worth releasing."[9] Supporting cases fill the ledgers: ten review rounds that "did not add substance"; a snapshot that changed only title metadata yet passed its checkpoint; a review loop that became "recursively self-referential"; a counting system that the agent satisfied by atomizing six packets into 150 countable rows; and the closing admission that "the repository can describe the failure more accurately than it can prevent the production behavior." Reviewer contamination is documented too: agent reviewers who shared repository context accepted documents that the real reader then rejected, while low-context persona reviews with forced-choice questions ("would you act now?") were the one automated signal that predicted the human verdict.

2.3 SMEwallet: the reader gate that worked

SMEwallet's north star went through at least eleven drafts. Draft 10 was rejected wholesale as the readable artifact; its hashes are frozen as "the rejected scientific baseline," and the Draft 11 program plans against the failure modes that produced it, by name: wrong baseline, "brevity as a proxy," table density, technical-dump risk.[10] Draft 11 rebuilt the document as ten storyboarded

[6] `DynareMCP/docs/AIpostdoc/results/scholarly_writing_failure_root_cause_audit_2026_07_18.md`, tell H. The same audit's registry F01–F25 supplies the cases cited below.

[7] `DynareMCP/docs/AIpostdoc/finalBGS/archive/2026_08_12/bgs_whole_document_natural_prose_audit_2026_08_02/restart_v7/frozen_repair_map_v7.jsonl`, which also preserves all 815 before/after pairs.

[8] `DynareMCP/docs/AIpostdoc/reviews/full_reader_facing_proposal_round_2026_06_06/reader_pull_reviews/reader_pull_aggregation.md`.

[9] `DynareMCP/docs/AIpostdoc/results/ai_junior_pi_proposal_to_behavior_root_cause_audit_2026_07_18.md`.

[10] `SMEwallet/docs/plans/sme-wallet-north-star-draft11-linear-novel-reconstruction-program.md`, section 12.

narrative units, each of which had to pass a reader-reconstruction gate with the PI before the next was attempted. The handoff records read differently from anything in the BGS lane: "The PI's verdict was: 4, 5, 6 are good."[11]

Two principles from this effort carried into policy. First, difficulty is not the enemy: "Do not omit difficult details to make the document readable. Make the difficult details understandable." The Chapter 3 record shows what that means in practice: an eight-page candidate failed the PI gate for missing connective derivations, an eleven-page repair still left derivations implicit, and the accepted version was thirteen pages.[12] Length went up while readability went up; shortening is not the remedy. Second, protected-source maps: every chapter carries a ledger tying rewritten prose to the Draft 10 material it preserves, so aggressive rewriting cannot silently drop mathematics. The project's writing rules, now in its `AGENTS.md`, are the source of the linear-narrative skill described in Section 3. Its final status line is honest about the boundary: `human_peer_review_not_claimed`. Unit-level PI acceptance exists; whole-document acceptance was never claimed.

2.4 BayesFilter: five layers, uncovered in order

BayesFilter contains two writing histories. The long one is the p12–p50 lineage of companion notes reconstructing the Zhao–Cui tensor-train filtering papers: p13 was the first "human-readable" pass, driven by the observation that the source "still reads too much like an audit/proof artifact"; at p48 the user intervened because the note read "as several strong notes interleaved at once" and asked for a major reorganization; p49 was a voice pass; p50 attacked "chapter interior discipline," removing inventory lists and "any sentence that feels like it is already anticipating late-stage defense language."[13] Three full-document rewrites after the mathematics was already right. The whole-book version of the same story is the August monograph review, whose findings interleave substance (a confirmed square-root UKF factor error) with register ("phrases such as 'Phase 2B literature gate,' 'row-173 trace,' 'Lane B' . . . are unresolved for a standalone reader"), and which also catalogues a reviewer failure: verification bravado ("digit-exact," "every printed digit") reclassified on audit as reported, not reproduced.[14]

The short history is the ZLB survey rescue of 18–21 August, recorded for this project in the case study.[15] A 2,326-line internal markdown survey became a 49-page LaTeX manuscript through five editorial passes, each triggered by one round of owner feedback, and the owner's complaints arrived in a fixed order: the conversion was unreadable; the content was internal ("derived from the code inspected on 19 August 2026" means nothing to a reader); the style was mechanical (measured: 133 em dashes in 46 pages, 20 of 55 sections opening with "The," every section in one identical shape); the register was defensive ("reads like a lawyer protecting his client," roughly 65 flagged sentences, half rewritten); and only then, with the prose out of the way, a substantive gap surfaced (no named ladder of verification models). The case study's summary is the organizing insight of this whole document: "each layer of badness masked the next. Nobody can notice defensive register while the page is full of file paths."

[11] `SMEwallet/docs/handoffs/sme-wallet-north-star-draft11-batch1-reader-handoff.md`.

[12] `SMEwallet/docs/handoffs/sme-wallet-north-star-draft11-technical-reintegration-reboot.md`.

[13] The plans are under `BayesFilter/docs/plans/`, e.g. `p48-source-code-discrepancy-and-rewrite-master-plan-2026-06-12.md` and `p50-final-readability-assessment-2026-06-12.md`.

[14] `BayesFilter/docs/plans/bayesfilter-monograph-main-readonly-review-findings-2026-08-03.md`.

[15] `BayesFilter/docs/surveys/zlb_discontinuous_hmc/humanvoice_case_study.md`; the pass-by-pass record is the addenda of `hostile_review.md` in the same directory.

Three findings from the rescue matter for design. The uniform section shape was caused by our own style contract: “a template that guarantees a floor also imposes a ceiling.” The repairs that preserved trust were guarded by scriptable invariants: equation bodies byte-identical (109/109 on every pass), label and citation censuses, compile gates, leak sweeps run on the rendered text rather than the source. And the editorial test that separated legitimate qualification from lawyering was a single question: does this sentence give the reader a reason, or only protect the author?

2.5 MacroFinance: review at book scale

MacroFinance’s CIP monograph (roughly 47 chapters, about 665 pages) went through a book-wide review-and-rewrite campaign in August, leaving six staged comparison trees and seven full-document snapshots.[16] The findings separate the failure classes cleanly. Substance: a propagating sign-error cluster in the basis and forward-premium algebra, verified to specific lines. Style: “bold-lead paragraph monotony, bullet overuse in specific chapters rather than pervasive ‘AI voice’,” and “list scaffolding” that “deadens the teaching voice.” Register: an LLM-agent chapter written in “industry-white-paper” register, including measured-sounding figures for an unbuilt system (“ $\sim$1 ms on GPU” with no artifact behind it) and a worked example that violates the traceability property it illustrates. Overclaiming: a prediction upgraded to empirical fact in the conclusion. The rewrite program’s stated goal, “direct human prose rather than scaffold-heavy, list-heavy, or AI-sounding prose,” came with a ban list (“critical, crucial, comprehensive, powerful, dramatically, seamlessly”) and a genre question that generalizes: does the material “sound like an academic monograph rather than a product white paper?” The campaign also shows review discipline the BGS lane lacked: the opinion accepted the findings as “high-value triage input” but blocked execution until blocker-level repairs were adjudicated, and every tree carries a build gate recording the commands actually run.

2.6 What the record adds up to

Reading the five projects together, ten conclusions survive:

1. **Failures arrive in layers, and repair must follow the layer order.** Format, then register, then mechanical tells, then defensiveness, then substance. Readability audits surface content audits; a style-only pass will certify an unreadable document.

2. **Internal register is the biggest single defect, and it is caused by context.** Whatever vocabulary the author-agent holds in working memory while drafting leaks onto the page. Reducing the ceremony an agent must carry while writing is a register intervention, not a convenience.

3. **Defensive prose is induced by integrity governance.** Non-claims, prohibitions, and moralizing tics (“honest,” “silently”) are the shadow of anti-hallucination rules. The cure is relocation and conversion, not deletion: integrity apparatus lives in a sidecar, and prohibitions become mechanisms the reader can re-derive.

[16] `MacroFinance/docs/latex-papers/docs/fable-rewrite/`, especially `findings.md`, `opinion.md`, and `rewrite-program-v1.md`.

4. **Templates have ceilings.** Every quality floor we wrote (section shapes, checklists, minimum elements) was promptly satisfied without the quality, and imposed its own monotony. Pattern lists and contracts must be diagnostics subordinate to judgment, never generators of prose.

5. **Local repair does not compose.** Opening fixes, bridges, and boxes improve adjacent pages, not arguments. The unit of failure is the document's dependency graph. This is why the 526-page volume could absorb 815 paragraph repairs and still lack a recorded reader acceptance.

6. **Agent self-review cannot certify a human voice, and context-sharing reviewers are contaminated.** The only automated signal that predicted human rejection was low-context persona review with forced choices. The human read is the acceptance gate; everything else clears ground.

7. **Both failure directions exist.** Documents were rejected for being compressed logbook summaries and for being sprawling logbooks. "Make it shorter" and "make it longer" are both false remedies; the target is selection and recomposition with the evidence preserved in a separate layer.

8. **Two documents, not one.** The internal record (ledgers, audit anchors, identifiers) and the human manuscript must be separate artifacts with an explicit identifier map. Every attempt to make one document serve both masters produced the hybrid that readers rejected.

9. **Measured diagnostics changed outcomes; vibes did not.** The ZLB repairs moved because someone counted (133 dashes, 20/55 openers, 65 defensive sentences) and then verified the counts after repair. The BGS lane's unmeasured review rounds did not converge.

10. **Guard the baseline or lose trust.** Byte-identity of equations, label censuses, and explain-every-removal reports are what made aggressive prose editing safe. "Readability improvement is not established by shortening."

One warning belongs beside the list. The BGS project learned to describe its failures better than it learned to prevent them, and rewarding honest self-assessment created an exit ramp ("honest failure labels became a possible escape hatch"). `humanvoice` should therefore judge itself by documents accepted, not by the quality of its own post-mortems. This document included.

3 The tools we already have

The doctrine layer is in place and battle-tested; the measurement layer does not exist. That is the one-sentence inventory. In more detail:

Policy. The global policy's "Reader-Facing Scientific Prose" section is the constitution: a precedence order (mathematical correctness, then source fidelity, then domain meaning, then reader comprehension, then template regularity), a ban on software-workflow metaphor in exposition, positive-first rigor (state the assumption and its purpose, not the denial of a claim nobody made), and acceptance rules under which no linter, self-review, or clean build can certify naturalness.[17] Beside it sit the non-evasion vocabulary ("correct," "wrong relative to the stated target," "unsupported," "not checked," "heuristic only") and the literature-audit policy with its six ledgers and hostile-review gate.

Skills. Two writing skills encode the SMEwallet and ZLB lessons. `write-linear-scholarly-narrative` covers document architecture: freeze the reader contract, design the narrative spine before the contents, storyboard bounded units, teach in a stated order, validate by reader reconstruction. `write-natural-analytical-prose` covers the sentence layer: freeze a meaning ledger, give each sentence one epistemic job with calibrated verbs ("shows" is not a synonym for "suggests"), concrete actors before abstract containers, qualifications spent once at the point of likely misunderstanding. Both carry anti-template clauses learned from the ceiling problem, and both place human review as the only acceptance gate. A third home exists: MathDevMCP ships `repair-scholarly-readability`, the bounded-unit repair skill used on the industry-DSGE dossier, with the only working measurement scripts in the workspace (a surface auditor and a page-image renderer).

Diagnostics. The humanizer pattern list (vendored from `blader/humanizer` v2.11.2, current with upstream as of 20 August) is installed with a precedence header that demotes it to a diagnostic subordinate to the policy. That header matters: without it, a pattern list becomes a new template, and its flat em-dash rule would strip legitimate scholarly typography.

Infrastructure. `claudecodex` distributes all of this with an idempotent installer into the Codex and Claude configuration trees, wraps cross-agent review (probe ladders, bounded review packets, "VERDICT: AGREE or REVISE"), and carries a five-file proposal template system with seven evidence ledgers. Its tests, however, only assert that key phrases remain present in the skill files. Nothing tests whether an agent following a skill writes better.

Grounding tools. ResearchAssistant is the source-grounding half of citation discipline: local-first paper ingestion preferring arXiv LaTeX source, section and equation extraction, and a survey mode that implements the six literature-audit ledgers as software, with honest terminal states when discovery is blocked (as happened in the ZLB campaign). MathDevMCP is the mathematical-integrity half: SymPy-backed equality checks, derivation audits, and an experimental whole-document rigor audit that produced one applied patch in the ZLB manuscript. DynareMCP contributes structure inventories used to falsify inflated rewrite claims. All three respect the same boundary: they nominate and veto; they do not certify.

The gaps, then. None of the six tools the case study asked for exists as code: no register linter, no rhythm profiler, no defensive-register finder, no LaTeX-aware baseline differ, no first-class two-document convention, no policy-precedence mechanism beyond one hand-written header. The scholarly writing skills live in two repositories with overlapping scope. Skill installation targets Codex first. And the evaluation question (does following the skill measurably help?) has never been asked of any of it.

4 What the research literature says

A survey agent verified the sources below against arXiv, Crossref, and publisher metadata; we read abstracts and summaries, not full texts, so we cite them for orientation and for design guidance, not for theorem-level claims. Four threads carry the weight.

[17] `claudecodex/policies/global-scientific-coding-agent-policy.md`, "Reader-Facing Scientific Prose" and "Plain Scientific Language And Non-Evasion."

4.1 The fingerprint is distributional and structural

Kobak et al. [2025] anchor the measurement methodology: borrowing the excess-mortality design from epidemiology, they track word frequencies across fifteen million PubMed abstracts and find a sharp post-2023 excess in style words ("delves," "underscores," "pivotal") rather than content words. The lesson is not the word list; it is that the fingerprint is a *distributional shift measurable against a pre-LLM corpus in the target genre*, no detector required. Juzek and Ward [2025] trace the overrepresentation to the human-feedback stage of training. The Wikipedia catalogue maintained by WikiProject AI Cleanup [WikiProject AI Cleanup, 2026] is the best practitioner seed list of features to measure, with the explicit warning that no single sign is diagnostic. Shaib et al. [2025] ground "slop" as a construct in expert annotation (low information utility plus verbosity and formulaic phrasing), and homogenization is documented experimentally: co-writing with an instruction-tuned model reduces content diversity relative to both solo writing and base-model assistance [Padmakumar and He, 2024, Doshi and Hauser, 2024]. One further result reached us through an open-source project rather than the literature agent: discourse-level features alone reportedly detect machine narrative at 93% F1 while professional surface rewriting barely moves detection [Russell et al., 2026]. We have not inspected that paper; if it holds, it independently confirms our "local repair does not compose" lesson at the level of detection statistics.

4.2 Alignment training explains the register

These tells are not accidents of one model. RLHF measurably trades diversity for preference score [Kirk et al., 2024]; typicality bias in human preference data sharpens the output distribution toward the mode, and inference-time interventions can partially reverse it [Zhang et al., 2025]; reward models correlate strongly with length, which is the mechanistic root of padding [Singhal et al., 2023]; and preference training produces sycophancy, the same mechanism that rewards hedging and throat-clearing [Sharma et al., 2024]. Recursive training on synthetic text deepens all of it [Guo et al., 2023]. Two design consequences: prompts alone cannot fully undo a training-time artifact, so an editing pass with explicit constraints beats an appeal to the model's taste; and compression is the highest-yield edit class, because verbosity is a known reward artifact.

4.3 Detectors cannot be gates

Detection research runs from GLTR's token-rank visualization [Gehrmann et al., 2019] through probability-curvature methods [Mitchell et al., 2023, Bao et al., 2024] to paired-perplexity scores [Hans et al., 2024]. Its practical upshot for us is negative: Liang et al. [2023] found commercial detectors flagging more than 61% of real TOEFL essays by non-native speakers as machine text; Weber-Wulff et al. [2023] found none of fourteen tools reliable; and Krishna et al. [2023] showed a paraphrase pass defeats the curvature family wholesale. Formal, polished scholarly prose is close to the worst case for false positives. Detector *internals*, used as continuous smoothness meters on aggregate corpora, are the most that can be salvaged, and nothing of the kind may ever gate a document or imply an accusation.

4.4 What actually works, and how to judge it

Evidence most strongly supports an edit-based pipeline over one-shot "write like a human" generation. Chakrabarty et al. [2024a] had professional writers edit LLM paragraphs, yielding a validated taxonomy of seven idiosyncrasies and roughly eight thousand fine-grained edits (the LAMP corpus); targeted edits substantially closed the quality gap. On evaluation: LLM judges agree with humans often enough to be useful proxies [Zheng et al., 2023] but favor their own generations through self-recognition [Panickssery et al., 2024], favor verbosity unless length is controlled [Dubois et al., 2024], and rate creative prose far more generously than expert humans do [Chakrabarty et al., 2024b]. The discipline is mechanical: judge with a different model family than the generator, pairwise, position-randomized, length-controlled, one rubric dimension per call, and validate the proxy against blinded human A/B judgments. Two calibration points from the readability literature: human academic prose is itself a degrading target, with sentence complexity and jargon rising for a century [Plavén-Sigray et al., 2017], and jargon density predicts fewer citations [Martínez and Mammola, 2021], so "sound like the median abstract" is the wrong objective. Syllable-count formulas are nearly useless here; cohesion- and register-aware measures are the right conceptual template [Graesser et al., 2004, Crossley et al., 2022, Biber, 1988], each referenced to the target genre. Human-likeness itself is a dial with genre-dependent optima, not a maximand [Cheng et al., 2025].

4.5 Economics and publishing norms

Korinek [2023] treats LLM writing assistance as legitimate productivity technology for economists. Prevalence work estimates 7–17% of recent CS conference review text was substantially LLM-modified [Liang et al., 2024a,b], with AI-assisted reviews measurably shifting scores [Latona et al., 2024]; adoption in research writing skews toward non-native speakers and junior researchers, whose styles are converging [Lin and Zhu, 2025]; and undisclosed chatbot artifacts appear verbatim in published papers [Glynn, 2024]. Journal norms (Nature portfolio, Science) have settled on disclosure. The implication for humanvoice is a framing decision: the goal is prose that meets the author's own quality bar under disclosure norms, and detector evasion is a non-goal. Since evasion is trivially cheap anyway [Krishna et al., 2023], detector scores cannot even define success.

5 What open source offers

We cloned and read fourteen projects: the four named in the task and ten found by search.[18] Table 1 summarizes the verdicts; the prose that follows explains the ones that matter.

Of the four named projects, only humanizer survives contact with our requirements intact, and we already run it; a diff against upstream shows our vendored copy current, differing only by our precedence header. stop-slop is a sharp essay-voice skill whose distinctive patterns (false agency: "the data tells us"; negative listing; interrogative-opener crutches) are worth absorbing, but its flat bans (all adverbs, all passives, "two items beat three") would damage scholarly prose, and "the data tell us" is ordinary econometrics. taste-skill, despite its stars, is a frontend design pack with no prose skill in it; its one instructive detail is sociological, a maintainer note that the agent "historically ignored em-dash limits when phrased as 'use sparingly'," which is an argument for programmatic dash counting rather than prompt-level bans. writing-agent we reject outright: its framing is detector evasion (checks named "Watermark Stripping," "Non-Native

[18] Clones are pinned under `humanvoice/vendor/` with a manifest of commits; working copies remain in the session scratchpad. The evaluation read pattern lists, prompts, and code, not READMEs.

Project	Kind	Verdict	Reason in one line
blader/humanizer	skill	adopted	already vendored; current with upstream v2.11.2
tbhb/vale-ai-tells + Vale	linter rules	adopt	111 rules + structural metrics as counts; needs curation
seyedehsanhadi/sloptrim	scorer	adopt	deterministic 0–100 score, LaTeX-tolerant, hook-ready
AIScientists-Dev/academic-humanizer	skill	adapt	the academic layer ours lacks; merge, do not install
sam-paech/slop-forensics	corpus tool	adapt	derives tell lists from our own output; needs econ baseline
baoguangsheng/fast-detect-gpt	detector	adapt	aggregate trend metric only, never a gate
conorbronsdon/avoid-ai-writing	skill + code	mine	tiered word table, injection boundary, FP corpus
hardikpandya/stop-slop	skill	mine	false agency, negative listing; flat bans unfit
NulightJens/humanizer-stack	skills + code	mine	structural-pass thesis; genre mismatch
sylvainhalle/textidote	LaTeX QA	pilot	the only true LaTeX-aware prose checker
textlint (+ latex2e plugin)	linter host	pilot	real LaTeX AST, custom rules in JS
amperser/proselint	linter	skip	pre-LLM content, no custom-rule mechanism
leonxlnx/taste-skill	UI skills	skip	frontend design pack; no prose content
itallstartedwithaidea/writing-agent	framework	skip	detector-evasion framing; a third of its checks are vacuous

Table 1: Open-source evaluation verdicts. "Mine" means lift specific patterns into our own catalogue; "adapt" means use with modifications; "pilot" means worth a trial in the LaTeX pipeline.

Bias Exploit"), which conflicts with our positioning, and its engineering is weak (roughly a third of its forty checks unconditionally pass).

More valuable than the shortlist are the finds. Three deserve adoption. *vale-ai-tells* is a professionally engineered Vale rule package (111 prose rules with a false-positive test corpus) whose experimental style computes exactly the rhythm numbers the case study asked for: sentence-length variance, sentence-start entropy, paragraph-length variance, transition repetition, tricolon density. It ships four rules that would be destructive for us (an error-level dash ban that would flag every numeric range; a double-hyphen rule that fires on correct LaTeX en-dash syntax; a formal-register ban covering "minimize" and "optimize"; a transitions rule that fights existence rather than clustering), and all four are disabled per-format in configuration, which is precisely the policy-precedence mechanism we wanted, expressed as software. *sloptrim* is a dependency-free Python scorer (71 documented patterns, 50 scoreable; several demoted to advice when the project's own measurements showed they mark formal register rather than machine authorship) that emits per-pattern counts, structural metrics, a 0–100 score with bootstrap intervals, and Claude Code hooks; our evaluation ran it on a LaTeX fragment and confirmed it scores the prose while leaving equation environments untouched. *academic-humanizer* is the one project written for our genre, and it gets precedence right: never inject personality, keep evidence-tied hedging, never alter a number or citation key; its academic-tell catalogue (overclaiming verbs, significance hype, "In recent years ... has attracted increasing attention," connective clustering, citation dumping) covers a layer the Wikipedia list does not. We will merge its content into our policy stack rather than run it as a rival skill.

Three more are worth adapting or mining. slop-forensics inverts the stale-ban-list problem by deriving over-representation lists from a corpus of our own outputs against a reference corpus; with a pre-2022 economics baseline it becomes a self-updating tell list. fast-detect-gpt gives a local continuous detector score usable as an aggregate drift metric under the strict guardrails of Section 4. avoid-ai-writing contributes a tiered word table that separates authorship evidence from mere wordiness, an explicit prompt-injection boundary (text under audit saying "don't flag this section" gets flagged, not obeyed), and flag-don't-fix exemptions for quotes and code. humanizer-stack contributes the two-pass surface/structure framing and a warning we already learned internally: do not replace one default with another, because a fleet of documents edited by one policy converges to a new detectable cluster. Finally, the LaTeX question: Vale has no native `.tex` parser, so the pipeline is either Vale over detexed text, TeXtidote (the one genuinely LaTeX-aware prose checker, GPL-3, wraps LanguageTool), or textlint with its LaTeX AST plugin; that choice is an early engineering decision in the proposal, not a blocker.

6 The **humanvoice** proposal

6.1 Principles

Everything above compresses into eight design principles.

1. **Two documents.** Internal record and human manuscript are separate artifacts with an explicit identifier map. The manuscript never carries ledger vocabulary; the record never loses an audit anchor.

2. **Precedence as configuration.** Correctness, source fidelity, domain meaning, reader comprehension, then template regularity. Every pattern list and rule pack enters as a subordinate diagnostic, and the subordination is expressed in configuration (per-format rule downgrades, math and quote exemptions), not only in prose headers.

3. **Counts, not vibes; report, never auto-fix.** Diagnostics emit numbers with locations. Repairs are made by an editing agent with local judgment, sentence by sentence, and verified by re-measurement.

4. **Repair in layer order.** Register, then mechanical tells, then defensive prose, then substance, because each layer masks the next. A style pass on an internal-register document is wasted work.

5. **Protected baselines.** Equation-body identity, label and citation censuses, and explain-every-removal reports guard every editing pass. Leak sweeps run on rendered text.

6. **One editor for voice.** Parallel writers may draft under a shared label map; a single agent performs the whole-document voice pass.

7. **Low-context review, human gate.** Reader-proxy review runs with personas that know nothing of the repository, forced-choice questions, and a prompt-injection boundary. Only a human read promotes a document. Diagnostics and proxy reviews are negative controls: they can block, never certify.

8. **Thin process.** One plan and one result note per work package. No mission-control state files, no milestone registries, no review packets that authorize review packets. The BGS record is the justification; process vocabulary in the author's context is itself a register contaminant.

Evasion is a non-goal throughout: humanvoice improves prose under disclosure norms and never optimizes against a detector.

6.2 Shape of the repository

humanvoice consolidates into five trees. `policies/` becomes the single home for the writing doctrine: the prose policy, the two writing skills (moved from claudecodex, which keeps distribution), the humanizer pattern list, and a new scholarly layer merged from academic-humanizer's academic tells and our own additions (false agency with an econometrics carve-out, negative listing, the defensive-register catalogue from the ZLB and BGS records). `diagnostics/` is the measurement layer, a small CLI (working name `hv`) wrapping curated external tools plus our own checks. `corpus/` holds the evaluation assets: the 815 BGS before/after pairs, the ZLB pass snapshots with their measured counts, the SMEwallet draft-10/draft-11 units with recorded PI verdicts, and a pre-2022 economics reference corpus for excess-frequency statistics. `harness/` holds reader-proxy review templates (descended from the AIpostdoc reader-pull personas, the one automated signal that predicted human rejection) and the behavioral tests. `vendor/` pins the adopted external projects with a commit manifest.

Most of the case study's wishlist becomes configuration rather than new code inside the `hv` CLI:

- `hv leak`: register linter; project-configurable banned vocabulary, path, date, and identifier patterns; runs on rendered text.
- `hv rhythm`: dash density, opener distribution, section-shape clustering, transition counts, sentence-length variance and burstiness (sloptrim and the vale-ai-tells experimental metrics, curated).
- `hv defensive`: candidate-sentence finder for negative definitions, prohibitions without mechanisms, moralizing tics, and unraised objections, emitting each sentence with context and the editorial question ("reason or protection?") for an editing model.
- `hv diff`: LaTeX-aware baseline differ; equation-body identity, label, citation, and reference censuses; an explain-every-removal report.
- `hv gate`: compile gate plus rendered-page image sampling, since text extraction lies about layout.
- `hv drift`: corpus-level trend metrics (slop-forensics over our own output against the economics baseline; optionally a detector-internals smoothness score, aggregate only).

6.3 Work packages

WP1, consolidation. Move the writing doctrine into `humanvoice/policies/`, merge the academic layer, keep claudecodex as the installer with `humanvoice` as source of truth, add Claude-native skill installation beside the Codex path, and absorb or cross-reference MathDevMCP's repair skill so scholarly readability has one home. Acceptance: one install command produces both agent setups; the existing phrase-level regression tests pass against the new layout.

WP2, measurement layer. Build `hv` as above. The acceptance test is historical replay: run against the preserved ZLB pass 2 snapshot, `hv rhythm` must reproduce the recorded numbers (133 em dashes, 20/55 "The" openers), and against the final manuscript, zero and four respectively; `hv diff` must confirm 109/109 equation identity across the recorded passes. Tools that cannot reproduce the history do not ship.

WP3, evaluation corpus and behavioral tests. Assemble `corpus/` and answer, for the first time, whether the skills work: generate matched rewrites with and without each skill, score with `hv`, and judge pairwise with a cross-family, position-randomized, length-controlled model judge validated against the recorded human verdicts in the corpus. Deliverable: a one-page scorecard per skill. This also gives us regression protection against the convergence trap: cross-document similarity is measured alongside per-document quality.

WP4, workflow. Codify the editorial pipeline as one skill: two-document setup with identifier map, layer-ordered repair, one-editor voice pass, per-agent drop reports, protected-baseline checks at every pass, reader-proxy review, human gate. Most of the content exists in the style contract and the case study; the work is consolidation and the removal of everything ceremonial.

WP5, pilot. Apply the full pipeline to one live document (the next BayesFilter companion note, or one CIP monograph part), with the ZLB rescue as the benchmark: five feedback passes to acceptance. The pilot's question is whether the toolkit reduces the number of human-feedback rounds, which is the only metric the record justifies caring about.

WP1 and WP2 are independent and can start immediately; WP3 needs WP2; WP4 needs nothing but benefits from WP2; WP5 needs all of them. Each package gets one plan and one result note, and any package that begins generating governance artifacts beyond that pair is, by the BGS lesson, failing.

6.4 Risks, and what is not established

Four risks have known shapes. A new template ceiling: one toolkit editing every manuscript will converge the fleet toward a new detectable cluster; WP3's cross-document similarity metric exists to catch this, and every diagnostic reports rather than auto-fixes for the same reason. Gate ossification: any floor we enforce will be satisfied without the quality; diagnostics therefore remain negative controls, and promotion authority stays with the human read. Measurement theater: a project about writing tools can spend itself on its own instrumentation; the one-plan-one-note cap and the WP5 pilot metric are the controls. Detector misuse: any per-document authorship score would be wrong on exactly the writers detectors already harm; `hv drift` is corpus-aggregate only and carries no per-document mode.

What this proposal does not claim: that any tool here, ours or adopted, improves human acceptance. The record contains unit-level PI acceptances (SMEwallet) and one document accepted after five tool-assisted passes (ZLB), but no controlled comparison. WP3 and WP5 are designed to produce that evidence, and until they do, the toolkit's status is: measured, plausible, unproven. The literature coverage is abstract-level and cited for orientation; the StoryScope result reached us secondhand and awaits inspection; sloptrim's accuracy figures are its own benchmark, not ours.

7 Conclusion

The workspace spent three months learning that review machinery cannot write, one month learning that a reader gate can, and four days learning that the distance from a correct internal document to a readable public one is five ordered layers of repair. The doctrine that survived is written down and installed; what is missing is measurement, evaluation, and consolidation, and each has a concrete work package above. The immediate next actions are mechanical: stand up the `hv` skeleton against the ZLB replay test (WP2), and move the policies into their single home (WP1). The first real test of humanvoice is the pilot: one live manuscript, and a count of how many rounds of human feedback it takes to reach "good."

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